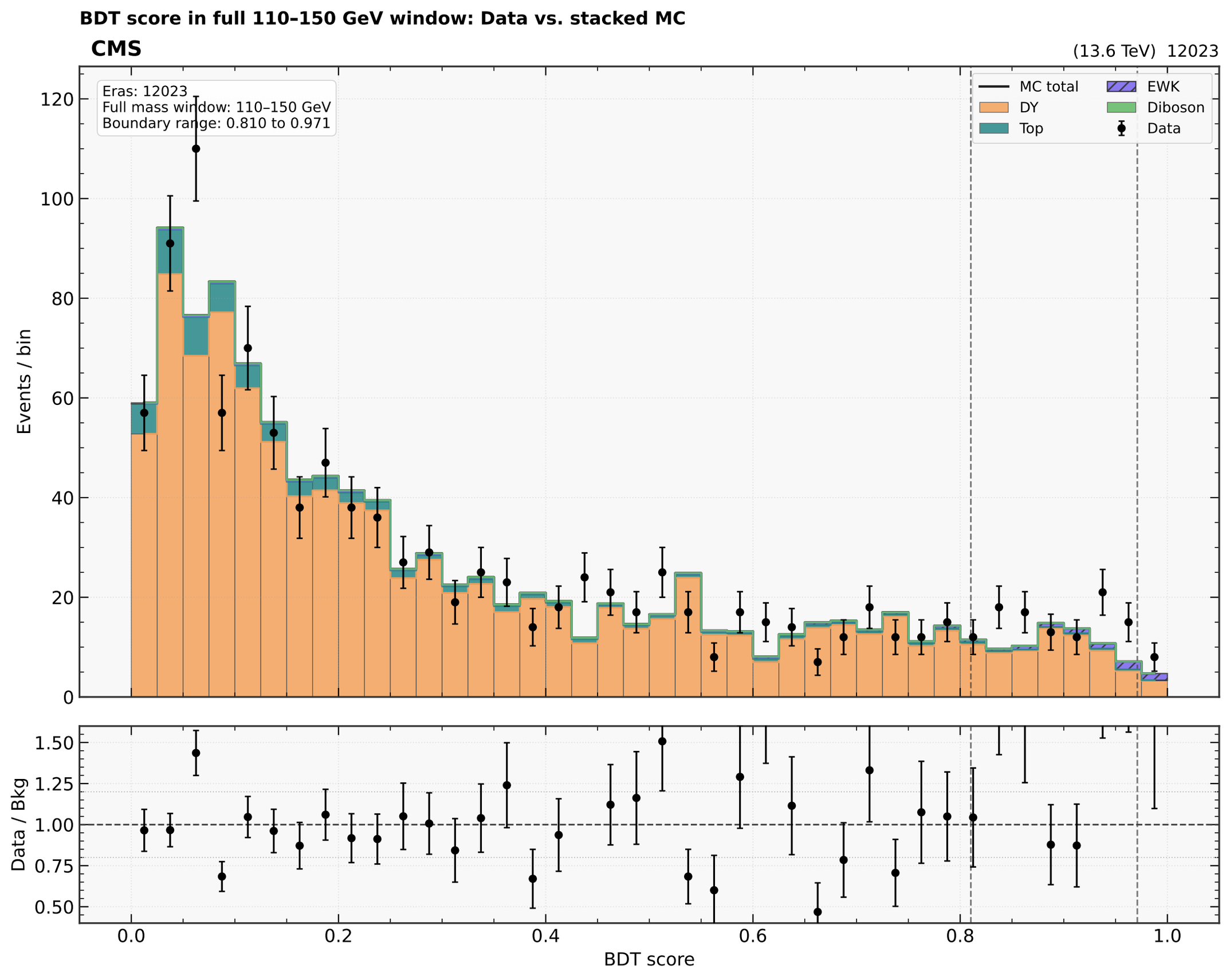
Run-3 H → μμ VBF BDT Category Optimisation

Eras : 12023
Category edges : 0.0000, 0.8103, 0.9500, 0.9710, 1.0000
MC scale factor : 1.0000 (Data/MC sideband normalisation)
Blinding window : 115–135 GeV
Sidebands : (110,115) ∪ (135,150) GeV
Outputs dir : /eos/home-m/memmad/HiggstoDimuon_VBF/src/drivers/9_TMVA/Outputs
Extra selection : –

CONTENTS:

p.2 BDT score distribution: Data vs. MC
p.3 BDT bin yields
p.4 Significance vs. number of categories
p.5 Boundary scan step-by-step (AN Fig. 32)
p.6 Atanh BDT + S/√B per category
p.7 BDT output vs. mass hypothesis (AN Fig. 26R)
p.8 Signal FWHM vs. BDT score
p.9 ROC curve with AUC
p.10 Blinded dimuon mass spectrum
p.11 Mass sculpting check (AN Fig. 25)
p.12 Per-category mass distributions and background fits
p.13 Background model comparison (χ^2/ndf)
p.14 VBF signal DCB fits per category
p.15 DCB parameter table
p.16 σ_{CB} vs. category
p.17 Signal acceptance × efficiency
p.18 Per-category yield and sensitivity summary
p.19 Signal mass shapes mH=120/125/130 (AN Fig. 40)
p.20 Category summary table
p.21 Category process composition

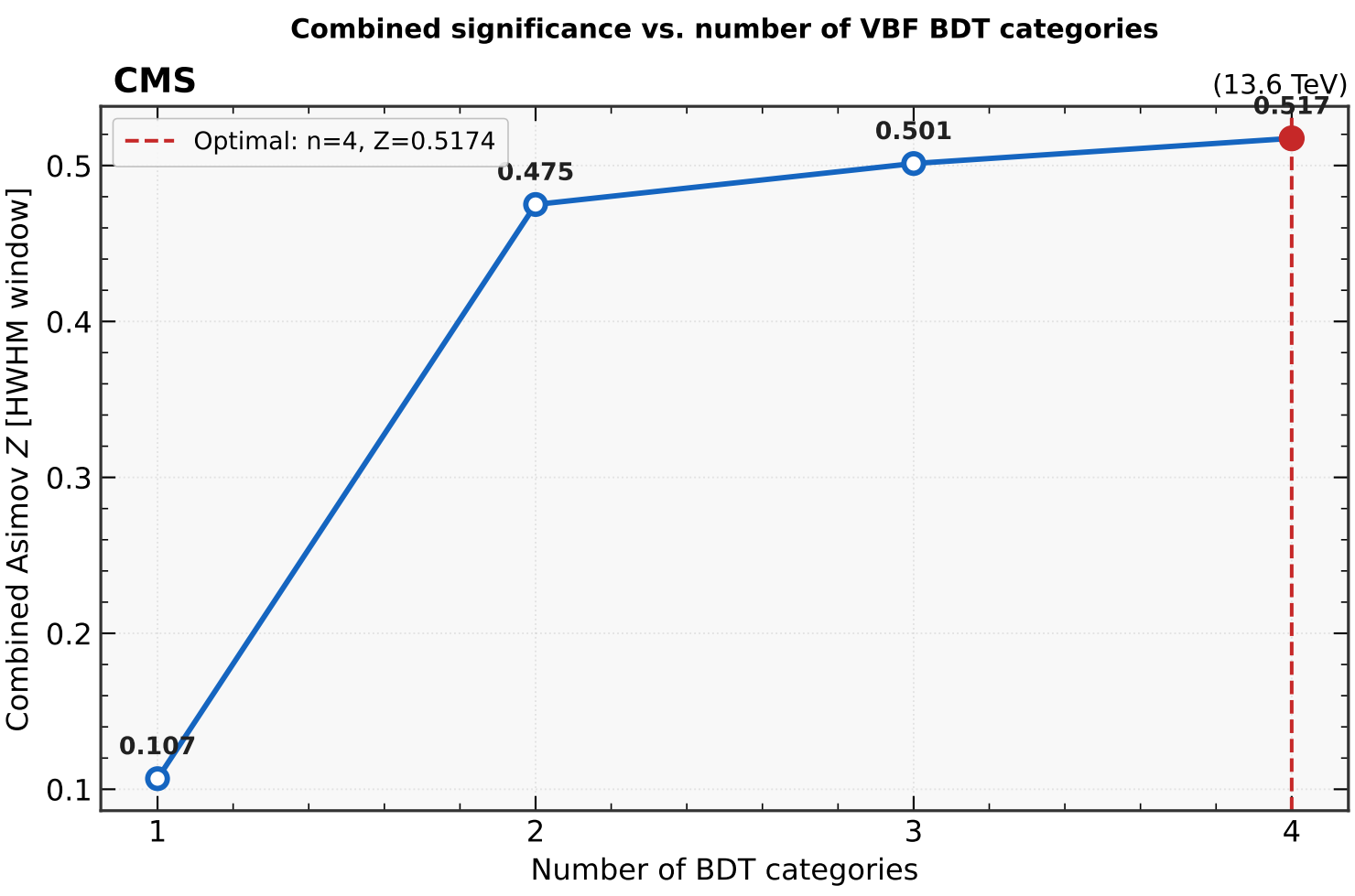

BDT score in full 110–150 GeV window: Data vs. stacked MC
CMS
(13.6 TeV) 12023
Eras: 12023
Full mass window: 110–150 GeV
Boundary range: 0.810 to 0.971
MC total
DY
Top
EWK
Diboson
Data
Events / bin
Data / Bkg
BDT score

BDT bin summary: signal, background, and sensitivity (page 1 / 2)

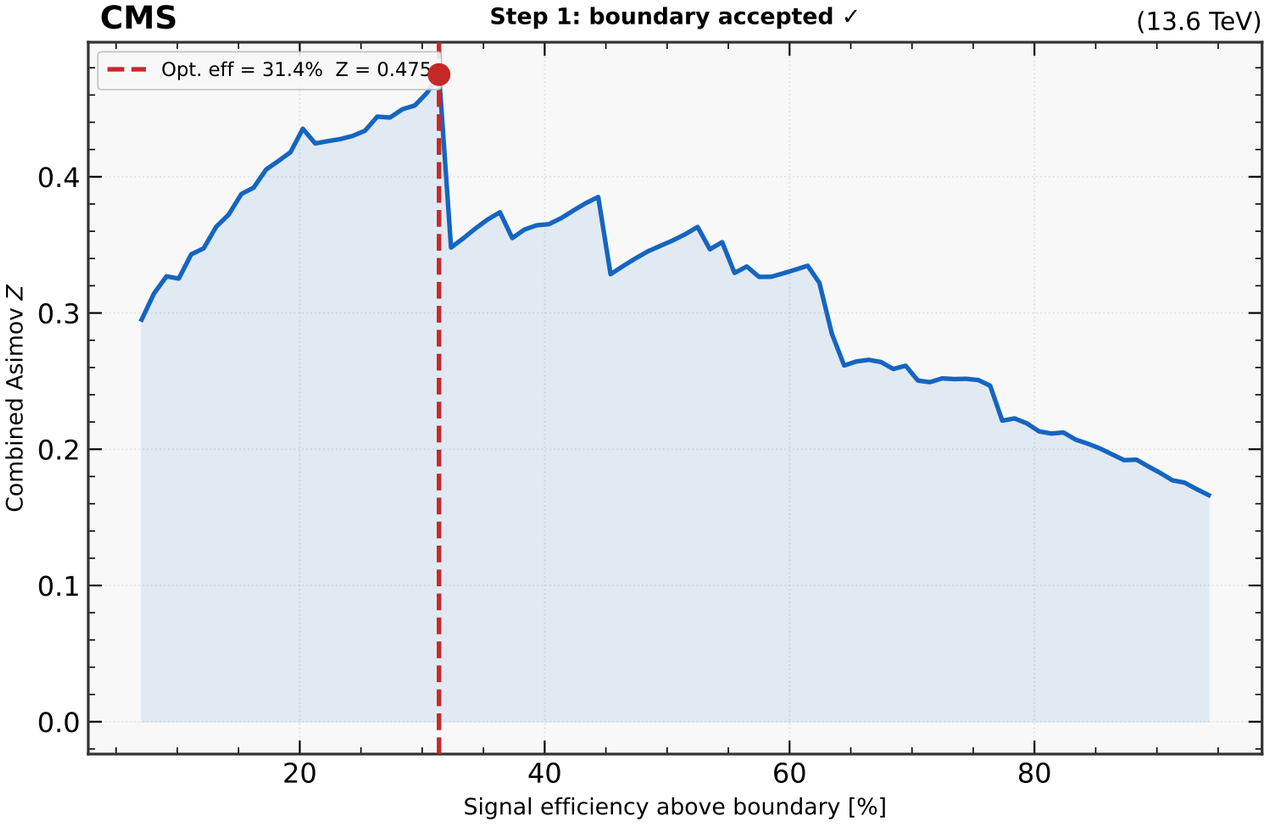
BDT bin	Total signal	Bkg. ± ΔB	Data	S/(S+B) (%)
[0.000, 0.025)	0.002	59.06 ± 5.15	57	0.00
[0.025, 0.050)	0.008	94.14 ± 6.63	91	0.01
[0.050, 0.075)	0.011	76.59 ± 5.85	110	0.01
[0.075, 0.100)	0.011	83.33 ± 6.38	57	0.01
[0.100, 0.125)	0.013	66.89 ± 5.80	70	0.02
[0.125, 0.150)	0.014	55.13 ± 5.19	53	0.03
[0.150, 0.175)	0.015	43.59 ± 4.67	38	0.03
[0.175, 0.200)	0.016	44.32 ± 4.69	47	0.04
[0.200, 0.225)	0.017	41.42 ± 4.53	38	0.04
[0.225, 0.250)	0.017	39.47 ± 4.62	36	0.04
[0.250, 0.275)	0.018	25.70 ± 3.64	27	0.07
[0.275, 0.300)	0.019	28.81 ± 3.67	29	0.06
[0.300, 0.325)	0.019	22.54 ± 3.25	19	0.09
[0.325, 0.350)	0.020	24.05 ± 3.41	25	0.08
[0.350, 0.375)	0.021	18.55 ± 2.84	23	0.11
[0.375, 0.400)	0.022	20.89 ± 3.42	14	0.10
[0.400, 0.425)	0.023	19.22 ± 3.13	18	0.12
[0.425, 0.450)	0.024	11.88 ± 2.21	24	0.20
[0.450, 0.475)	0.025	18.73 ± 3.08	21	0.13
[0.475, 0.500)	0.027	14.62 ± 2.79	17	0.18

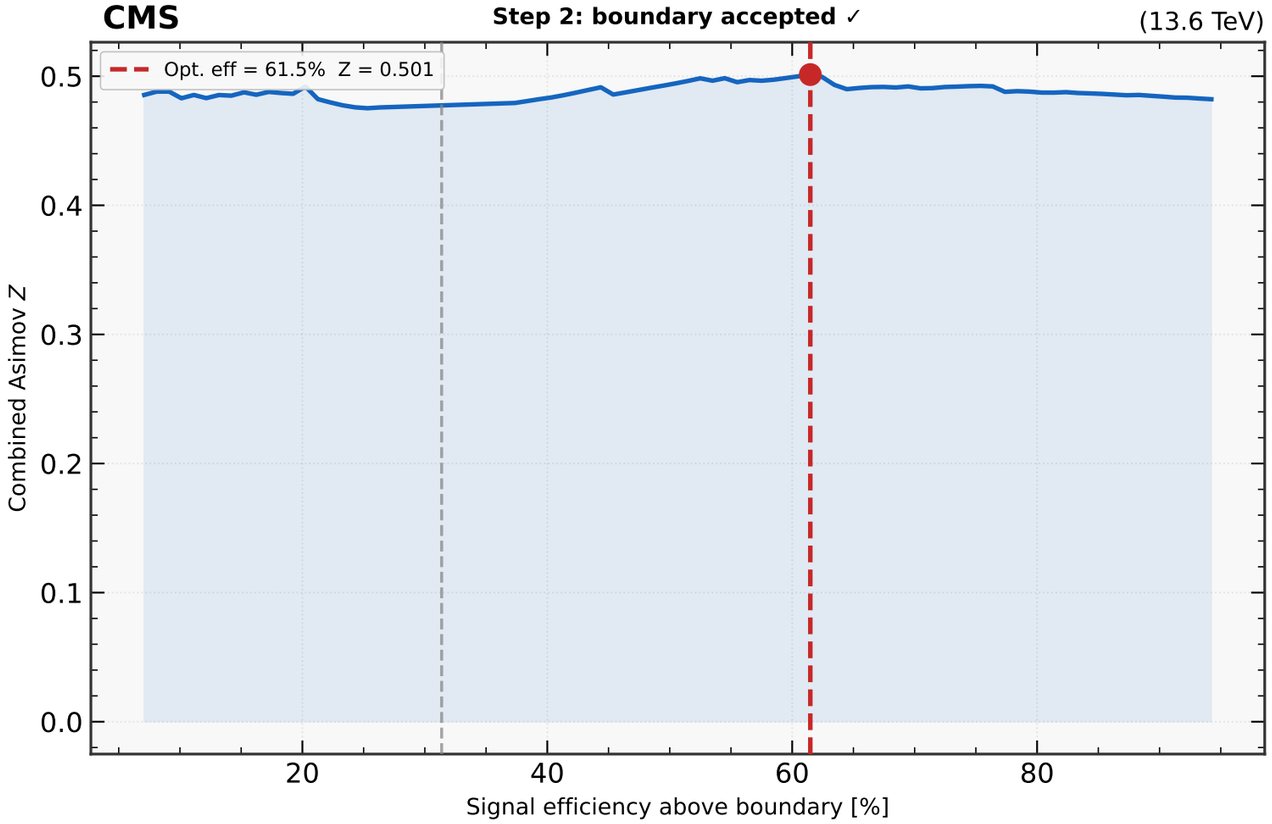
BDT bin summary: signal, background, and sensitivity (page 2 / 2)

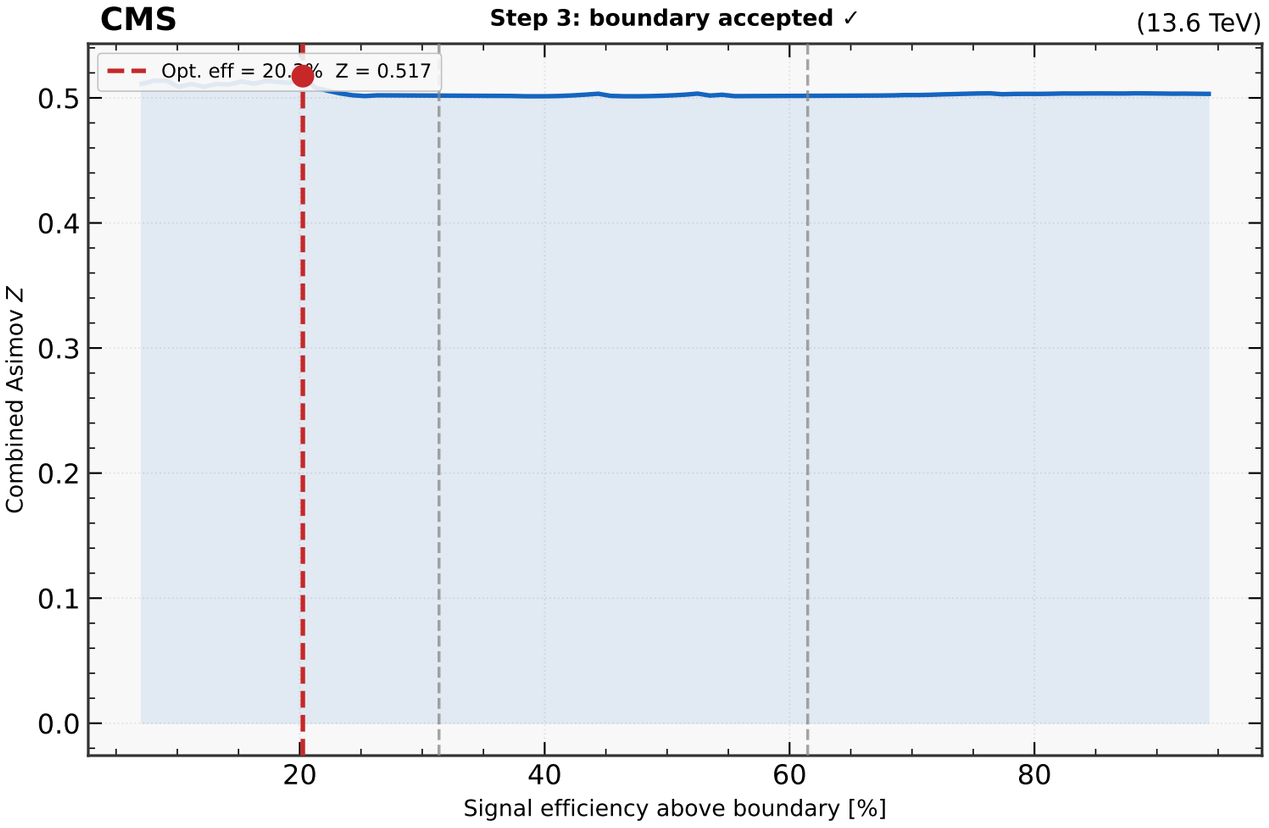
BDT bin	Total signal	Bkg. ± ΔB	Data	S/(S+B) (%)
[0.500, 0.525)	0.028	16.59 ± 2.90	25	0.17
[0.525, 0.550)	0.030	24.86 ± 3.59	17	0.12
[0.550, 0.575)	0.032	13.32 ± 2.43	8	0.24
[0.575, 0.600)	0.034	13.17 ± 2.43	17	0.26
[0.600, 0.625)	0.035	8.10 ± 1.88	15	0.43
[0.625, 0.650)	0.037	12.56 ± 2.44	14	0.30
[0.650, 0.675)	0.041	14.95 ± 2.63	7	0.28
[0.675, 0.700)	0.047	15.29 ± 2.77	12	0.31
[0.700, 0.725)	0.048	13.52 ± 2.57	18	0.35
[0.725, 0.750)	0.052	16.99 ± 3.06	12	0.31
[0.750, 0.775)	0.059	11.16 ± 2.18	12	0.52
[0.775, 0.800)	0.064	14.29 ± 2.67	15	0.45
[0.800, 0.825)	0.070	11.50 ± 2.31	12	0.61
[0.825, 0.850)	0.083	9.65 ± 2.19	18	0.85
[0.850, 0.875)	0.097	10.26 ± 2.23	17	0.93
[0.875, 0.900)	0.115	14.81 ± 2.70	13	0.77
[0.900, 0.925)	0.148	13.75 ± 2.54	12	1.07
[0.925, 0.950)	0.200	10.75 ± 2.22	21	1.83
[0.950, 0.975)	0.316	7.12 ± 1.50	15	4.26
[0.975, 1.000)	0.399	4.71 ± 1.32	8	7.82

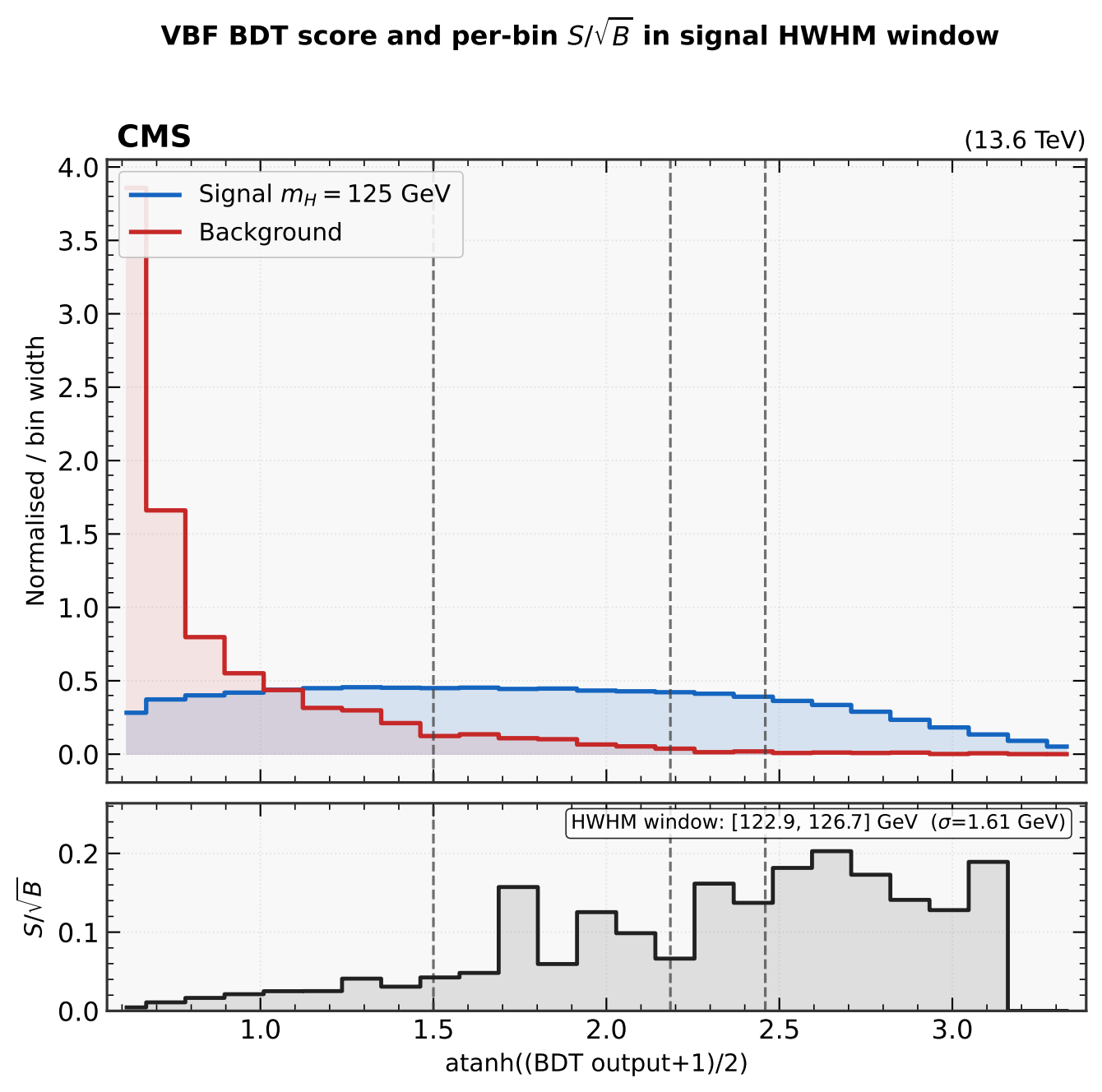

Combined significance vs. number of VBF BDT categories
CMS
(13.6 TeV)
Optimal: n=4, Z=0.5174
0.107
0.475
0.501
0.517
Combined Asimov Z [HWHM window]
0.1
0.2
0.3
0.4
0.5
1
2
3
4
Number of BDT categories

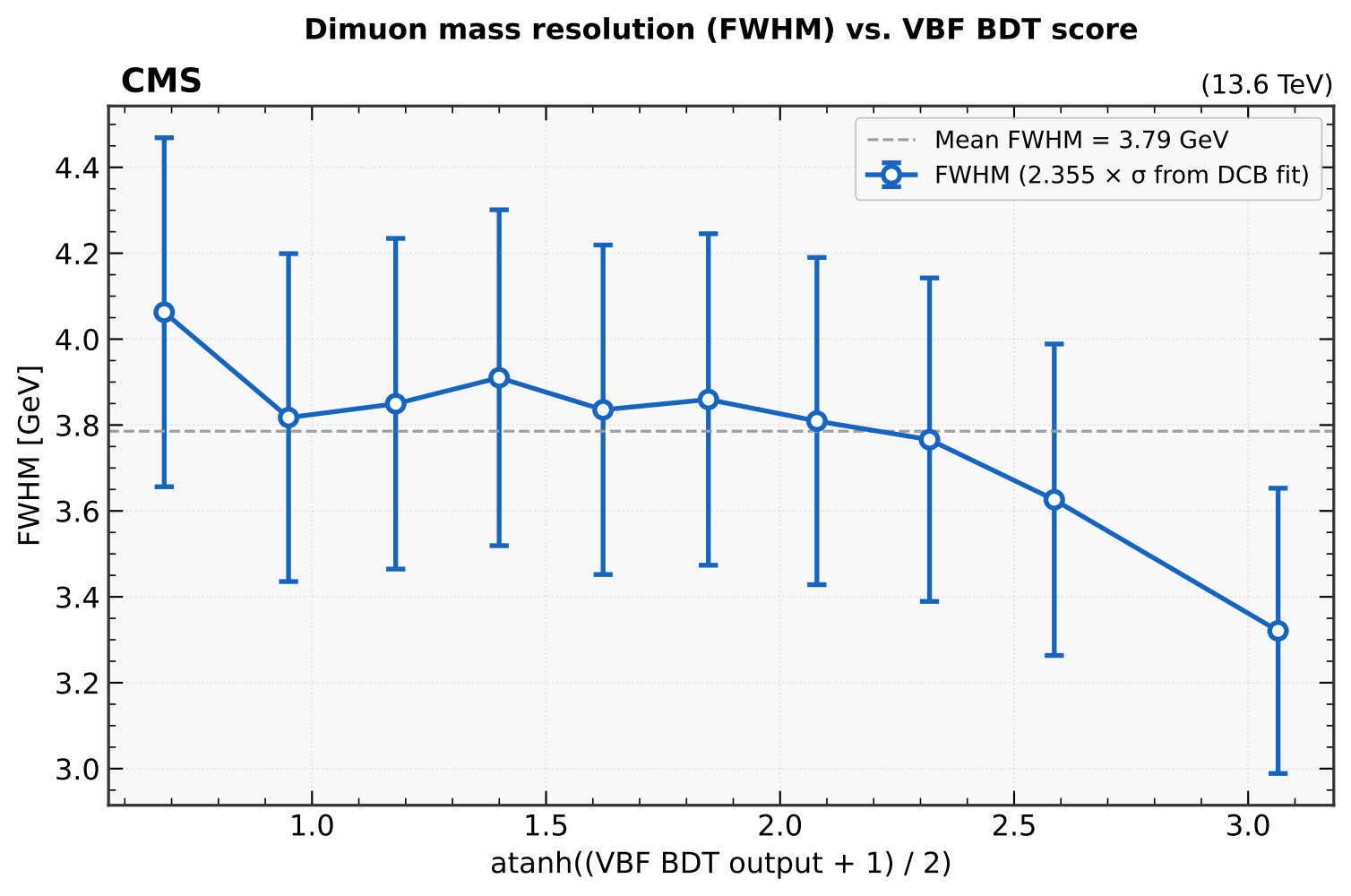
Iterative category boundary optimisation (AN Fig. 32 analog)

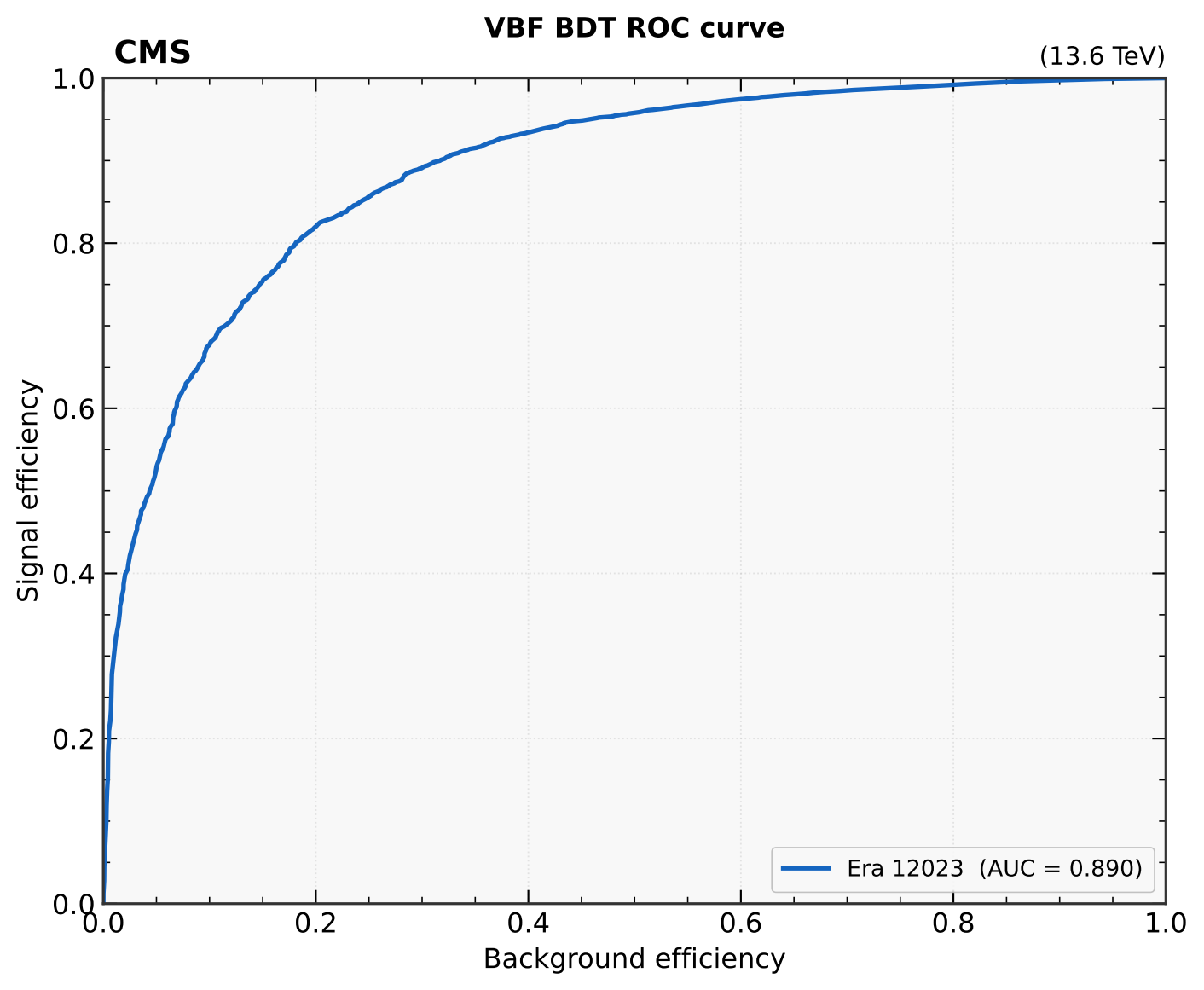


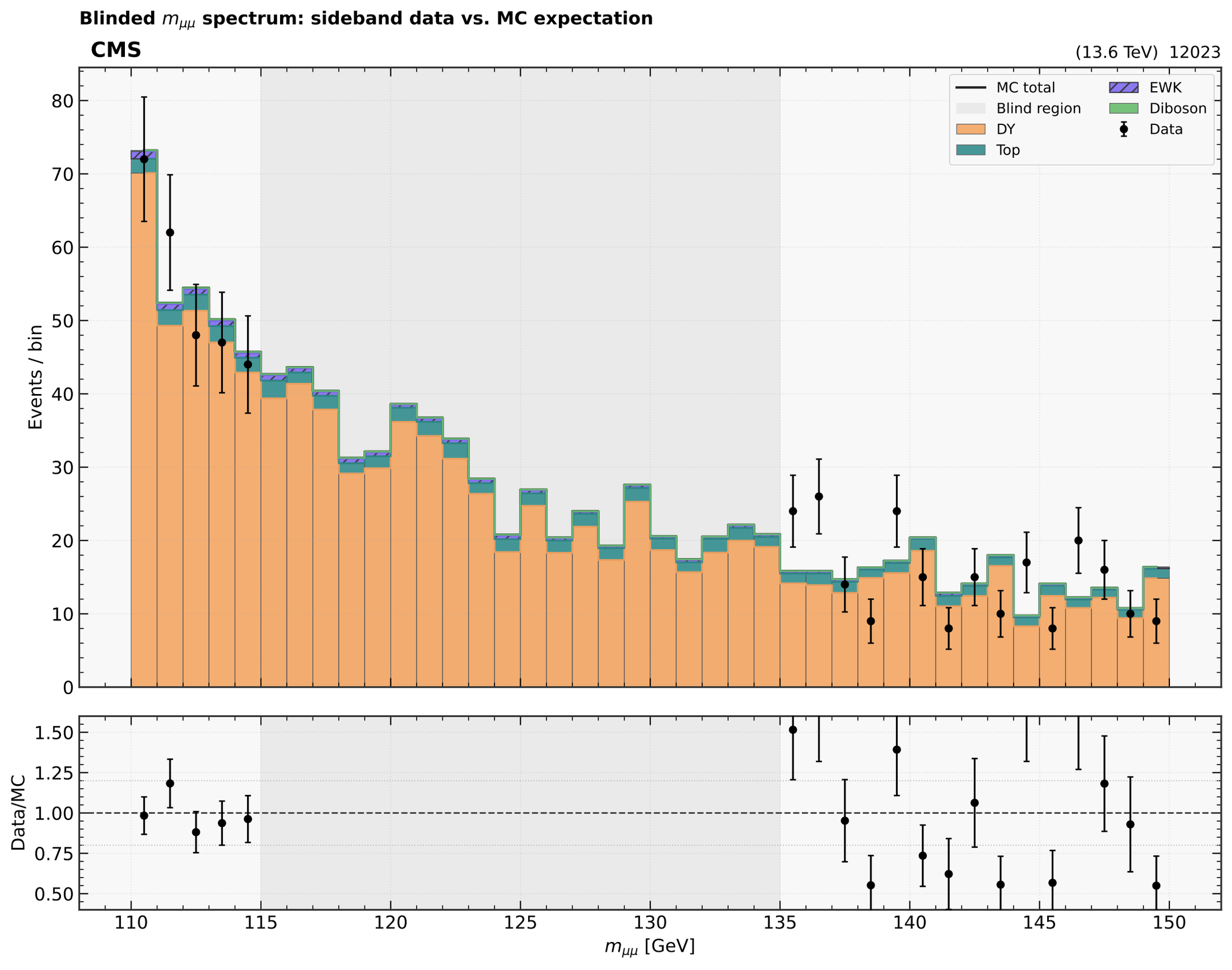


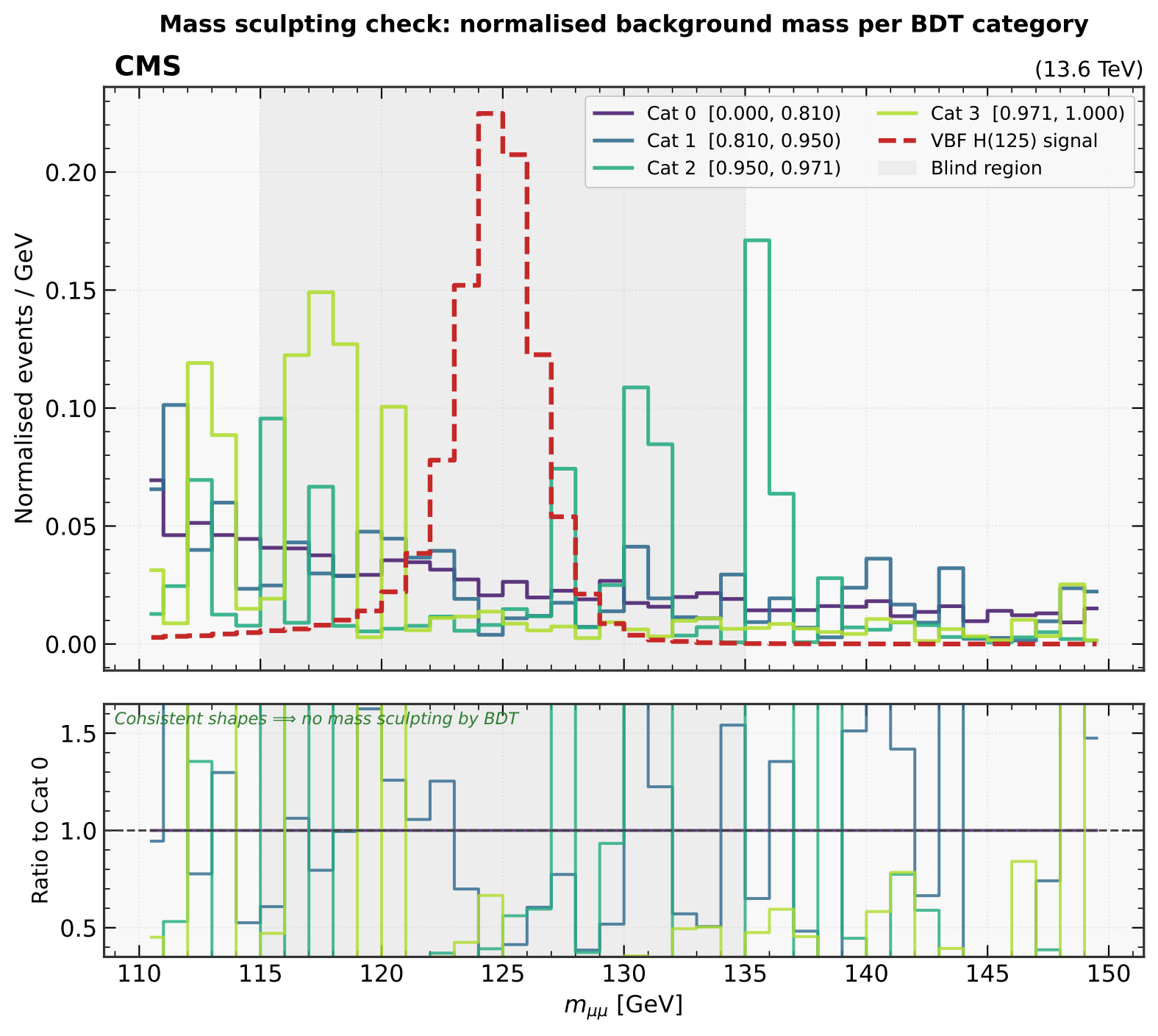



VBF BDT score and per-bin $S/\sqrt{B}$ in signal HWHM window
CMS
(13.6 TeV)
Signal $m_H = 125$ GeV
Background
Normalised / bin width
HWHM window: [122.9, 126.7] GeV (σ=1.61 GeV)
$S/\sqrt{B}$
atanh((BDT output+1)/2)

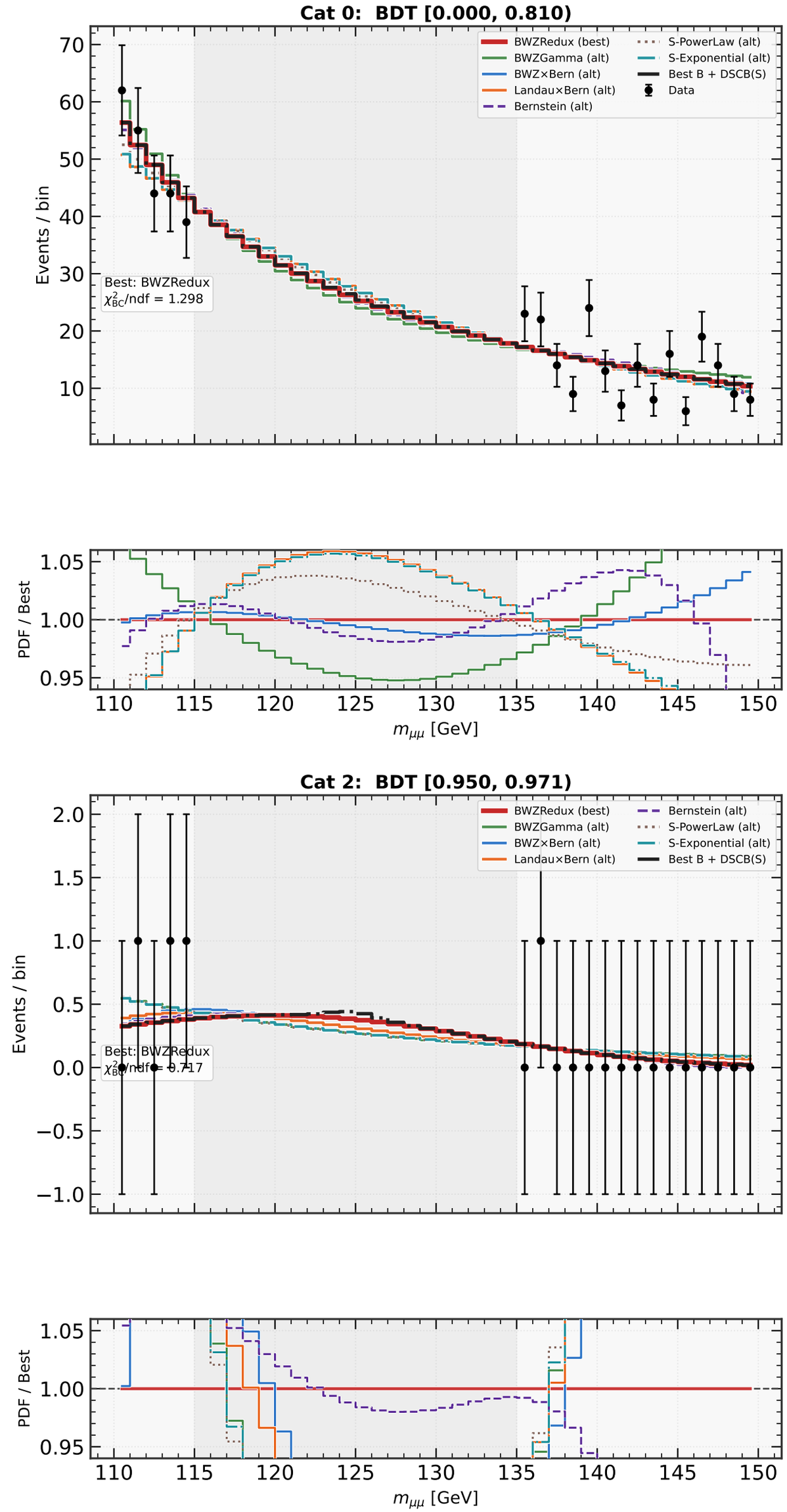

Dimuon mass resolution (FWHM) vs. VBF BDT score
CMS
(13.6 TeV)
Mean FWHM = 3.79 GeV
FWHM (2.355 × σ from DCB fit)
FWHM [GeV]
4.4
4.2
4.0
3.8
3.6
3.4
3.2
3.0
1.0
1.5
2.0
2.5
3.0
atanh((VBF BDT output + 1) / 2)

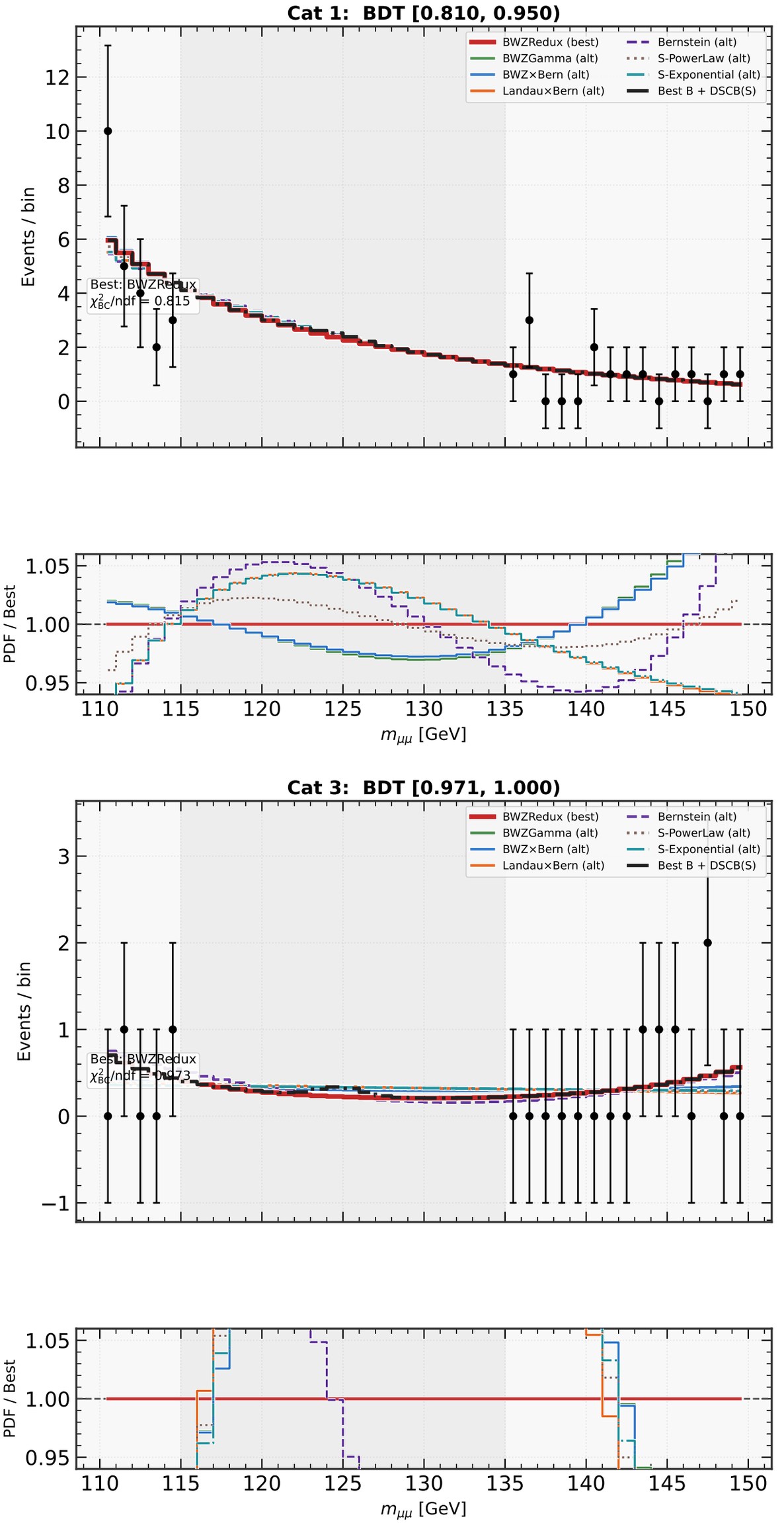

VBF BDT ROC curve
CMS
(13.6 TeV)
Signal efficiency
Background efficiency
0.0
0.2
0.4
0.6
0.8
1.0
Era 12023 (AUC = 0.890)


Blinded $m_{\mu\mu}$ spectrum: sideband data vs. MC expectation
CMS
(13.6 TeV) 12023
MC total
Blind region
DY
Top
EWK
Diboson
Data
Events / bin
Data/MC
$m_{\mu\mu}$ [GeV]


Mass sculpting check: normalised background mass per BDT category
CMS
(13.6 TeV)
Cat 0 [0.000, 0.810)
Cat 1 [0.810, 0.950)
Cat 2 [0.950, 0.971)
Cat 3 [0.971, 1.000)
VBF H(125) signal
Blind region
Normalised events / GeV
0.00
0.05
0.10
0.15
0.20
Consistent shapes ⟹ no mass sculpting by BDT
Ratio to Cat 0
0.5
1.0
1.5
110
115
120
125
130
135
140
145
150
$m_{\mu\mu}$ [GeV]

Per-category dimuon mass: background model fits

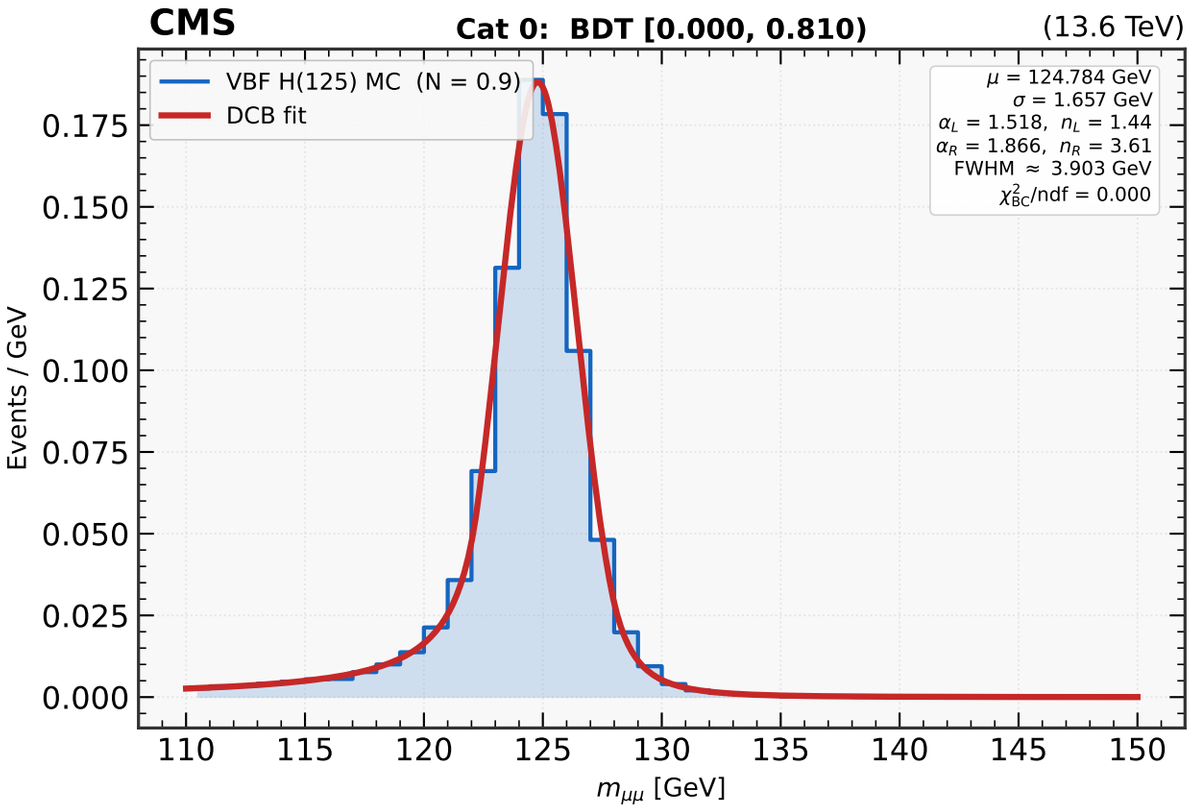


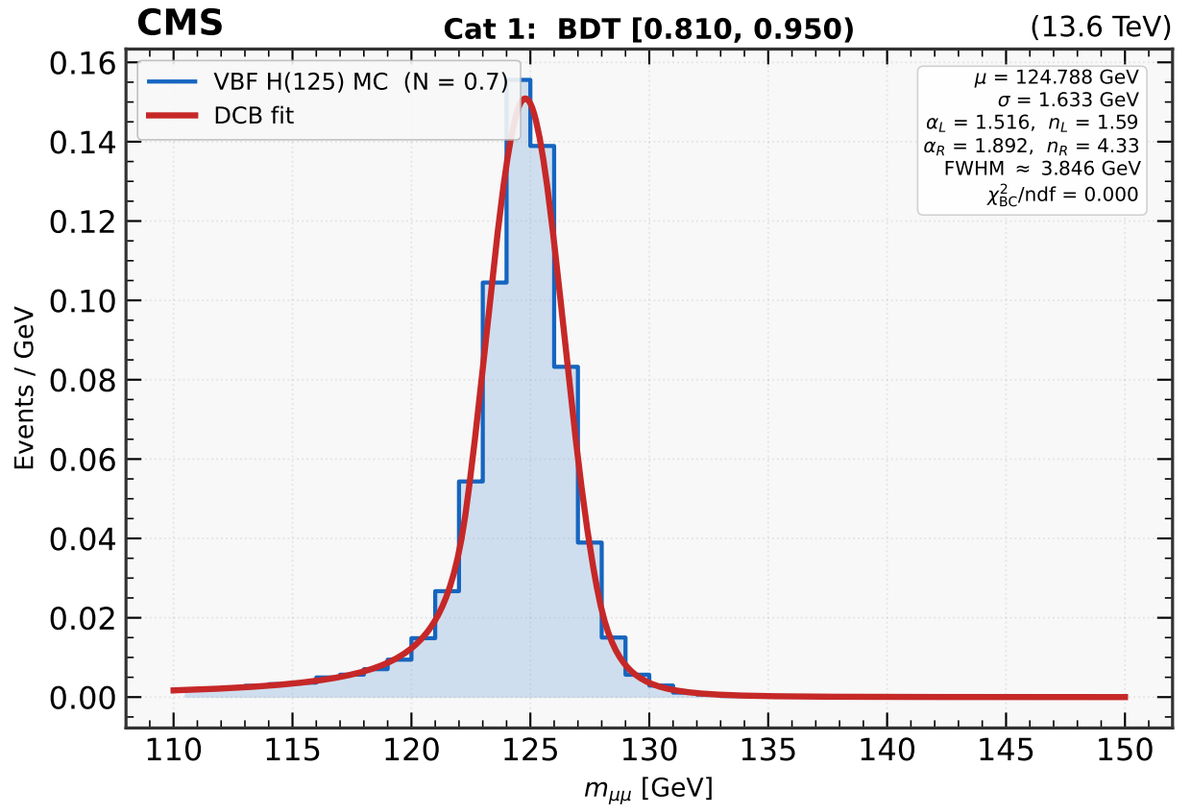


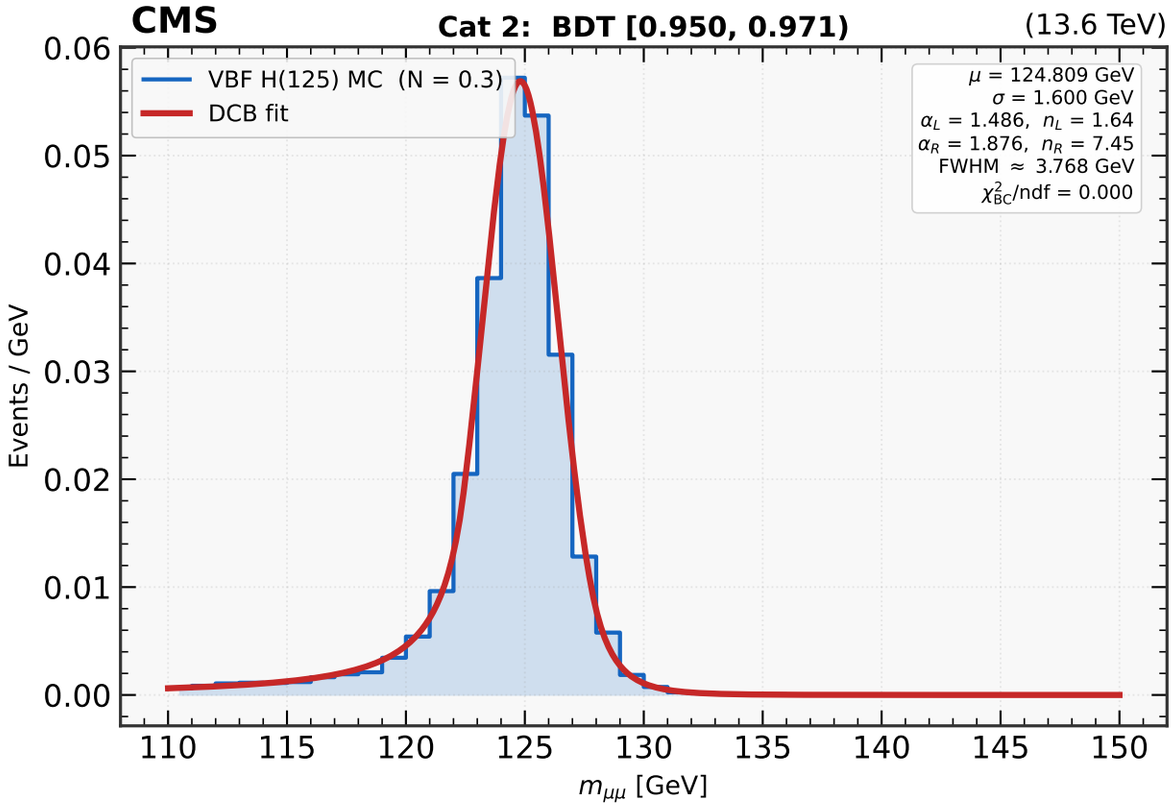
Per-category background model fit comparison (χ²_BC/ndf, ndf)

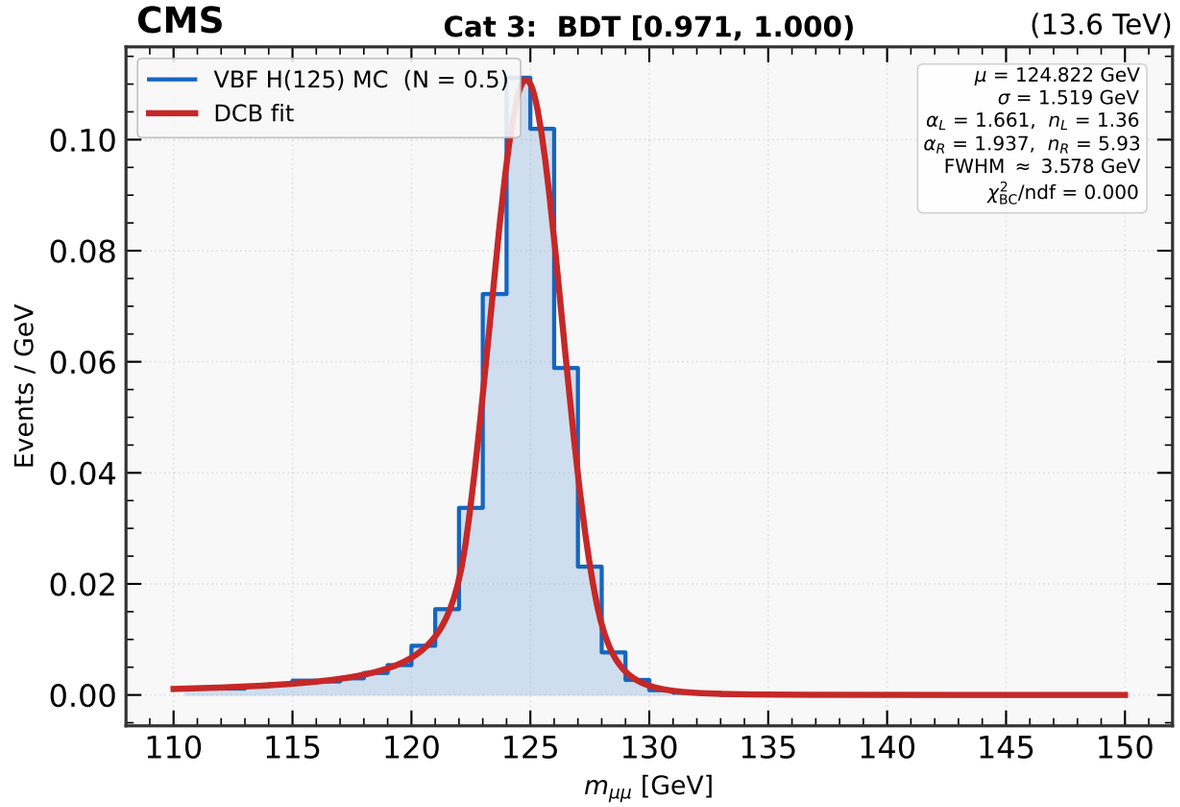
Category	BDT range	Best model	Best χ²_BC/ndf	BWZRedux	BWZGamma	BWZ×Bern	Landau×Bern	Bernstein	S-PowerLaw	S-Exponential
0	[0.000, 0.810)	BWZRedux	1.298	1.298 (37)	1.355 (37)	1.344 (36)	1.386 (37)	1.385 (35)	1.343 (37)	1.382 (37)
1	[0.810, 0.950)	BWZRedux	0.815	0.815 (37)	0.818 (37)	0.840 (36)	0.824 (37)	0.872 (35)	0.821 (37)	0.824 (37)
2	[0.950, 0.971)	BWZRedux	0.717	0.717 (37)	0.752 (37)	0.764 (36)	0.742 (37)	0.747 (35)	0.752 (37)	0.749 (37)
3	[0.971, 1.000)	BWZRedux	0.973	0.973 (37)	0.998 (37)	1.027 (36)	1.015 (37)	0.980 (35)	1.012 (37)	1.013 (37)

VBF H(125) signal DCB extraction per BDT category



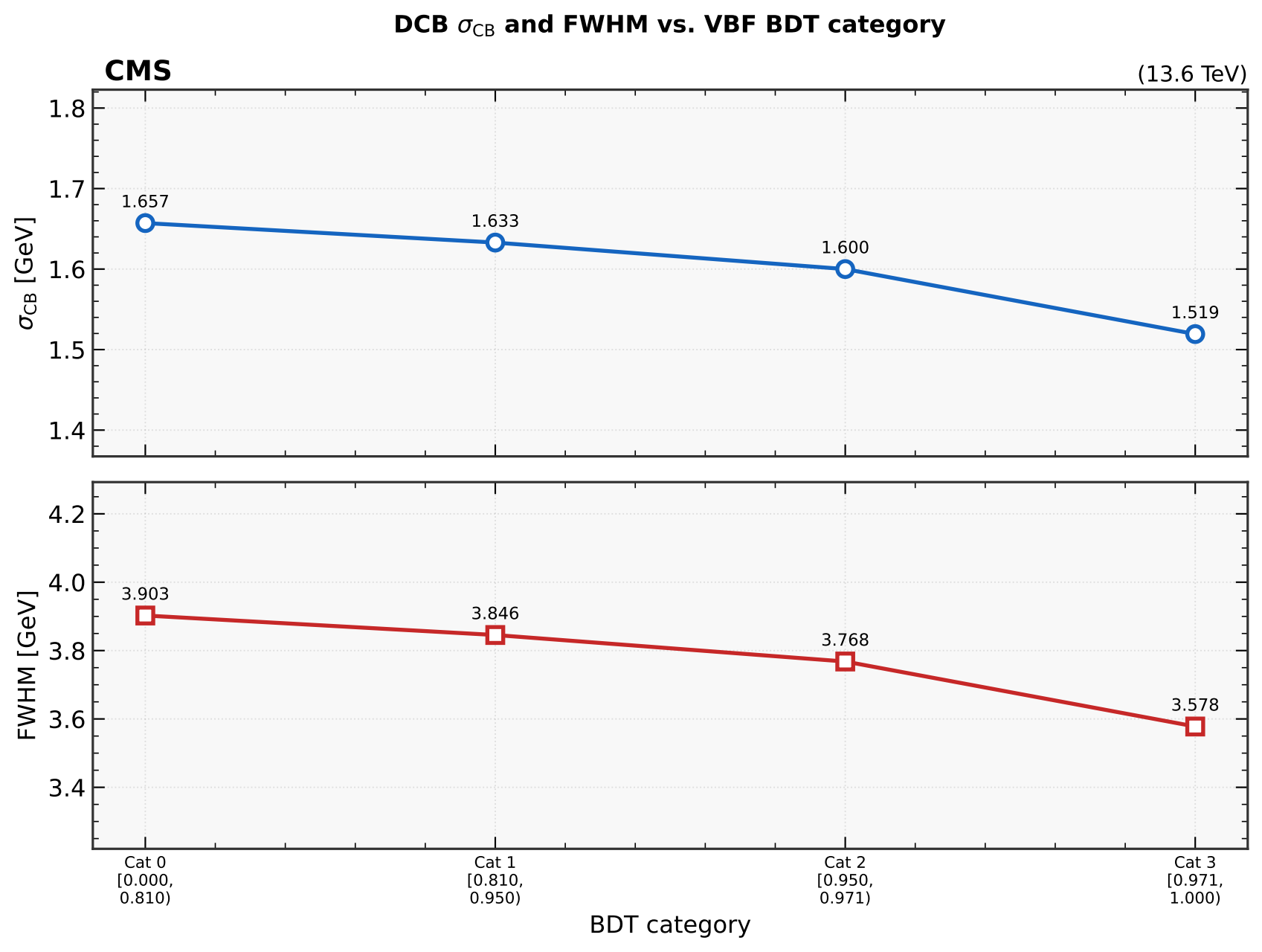


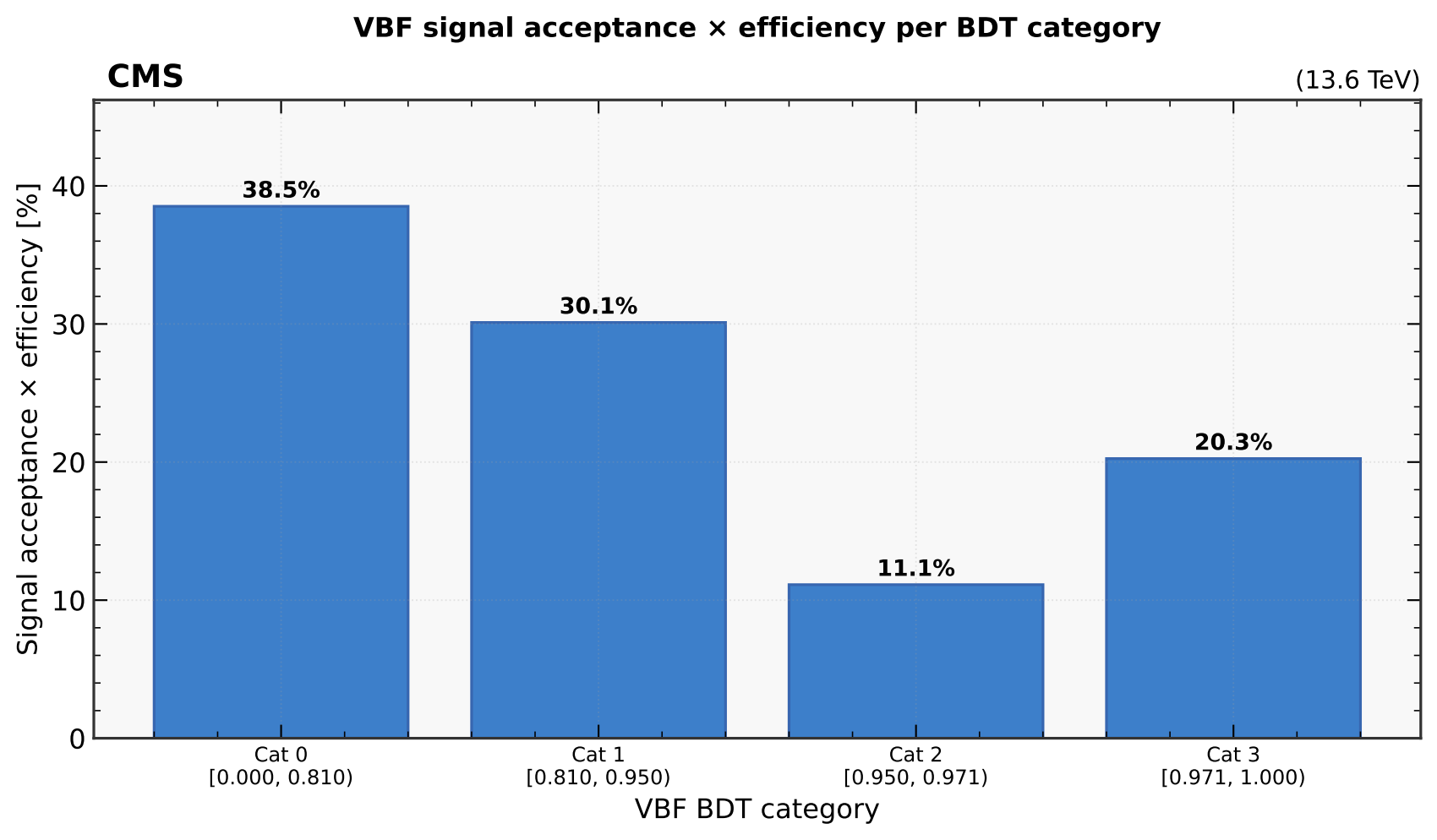




VBF signal DCB fit parameters per category

Category	BDT range	N_sig	Mean [GeV]	Sigma [GeV]	alpha_L	n_L	alpha_R	n_R	FWHM [GeV]	Asimov Z	χ^2_BC/ndf
0	[0.000, 0.810)	0.88	124.7836	1.6571	1.5181	1.435	1.8662	3.608	3.903	0.041	0.000
1	[0.810, 0.950)	0.69	124.7884	1.6329	1.5156	1.588	1.8922	4.327	3.846	0.172	0.000
2	[0.950, 0.971)	0.25	124.8085	1.6000	1.4863	1.638	1.8761	7.454	3.768	0.232	0.000
3	[0.971, 1.000)	0.46	124.8223	1.5193	1.6608	1.365	1.9374	5.933	3.578	0.420	0.000


DCB σ_{CB} and FWHM vs. VBF BDT category
CMS
(13.6 TeV)
σ_{CB} [GeV]
1.8
1.7
1.6
1.5
1.4
1.657
1.633
1.600
1.519
FWHM [GeV]
4.2
4.0
3.8
3.6
3.4
3.903
3.846
3.768
3.578
Cat 0 [0.000, 0.810)
Cat 1 [0.810, 0.950)
Cat 2 [0.950, 0.971)
Cat 3 [0.971, 1.000)
BDT category


VBF signal acceptance × efficiency per BDT category
CMS
(13.6 TeV)
Signal acceptance × efficiency [%]
0
10
20
30
40
38.5%
30.1%
11.1%
20.3%
Cat 0
[0.000, 0.810)
Cat 1
[0.810, 0.950)
Cat 2
[0.950, 0.971)
Cat 3
[0.971, 1.000)
VBF BDT category

Per-category yield and sensitivity summary

Category	BDT range	S_full	B_full	S_HWHM	B_HWHM	HWHM window [GeV]	S/√B	Asimov Z
0	[0.000, 0.810)	0.88	2314.44	0.599	210.880	[122.8, 126.7]	0.041	0.041
1	[0.810, 0.950)	0.69	150.69	0.472	7.377	[122.9, 126.7]	0.174	0.172
2 ★	[0.950, 0.971)	0.25	14.58	0.176	0.517	[122.9, 126.7]	0.245	0.232
3 ★	[0.971, 1.000)	0.46	13.07	0.325	0.501	[123.0, 126.6]	0.460	0.420

Category summary: HWHM-windowed signal and background yields

Category	BDT range	Data sidebands	MC sidebands ± stat	Data/MC (sidebands)	MC full (110-150) ± stat	S/√B [HWHM]	Asimov Z (cat) [HWHM]	B_HWHM
0	[0.000, 0.810)	450	1076.48 ± 34.93	0.418	2314.44 ± 51.25	0.0413	0.0413	210.880
1	[0.810, 0.950)	37	76.59 ± 9.39	0.483	150.69 ± 12.93	0.1739	0.1721	7.377
2 ★	[0.950, 0.971)	4	6.39 ± 2.13	0.626	14.58 ± 3.26	0.2447	0.2325	0.517
3 ★	[0.971, 1.000)	7	4.79 ± 1.73	1.461	13.07 ± 3.35	0.4595	0.4198	0.501
Combined	all						0.5115	

Category composition by process (full mass window)

Category	BDT range	DY	EWK	Top	Diboson	total	total_err2
0	[0.000, 0.810)	2139.69	22.50	146.21	6.04	2314.44	2626.87
1	[0.810, 0.950)	135.53	9.73	5.04	0.39	150.69	167.29
2	[0.950, 0.971)	10.87	3.01	0.62	0.08	14.58	10.66
3	[0.971, 1.000)	8.88	3.76	0.40	0.03	13.07	11.23